

EXPERIMENT 14:

The screenshot displays an IDE with the following components:

- Registers Window:** Shows the current state of the microcontroller's registers. The registers are organized into groups: R0-R7, Sys, A, B, SP, MAX, DFR, PC, Status, and PCSR. The values are as follows:

Register	Value
R0	0x09
R1	0x39
R2	0x0c
R3	0x0f
R4	0x00
R5	0x00
R6	0x00
R7	0x10
Sys	0x00
A	0x01
B	0x21
SP	0x27
MAX	0x0c39
DFR	0x0c27
PC	130691
Status	0x0034550
PCSR	0x00
- Code Editor:** Shows the C program `serial.c`. The code includes `<reg51.h>` and `<stdio.h>`. It defines `void main (void)` which configures Timer 1 (TMR1) and UART, then enters a loop printing "Hello World".

```
1 #include <reg51.h>
2 #include <stdio.h>
3
4 void main (void)
5 {
6     SCON = 0x50; /* SCON: mode 1, 8-bit UART, enable rxvr */
7     TMR1 = 0x20; /* TMR1: timer 1, mode 2, 8-bit reload */
8     TH1 = 0xf0; /* TH1: reload value for 9600 baudrate */
9     TR1 = 1; /* TR1: timer 1 run */
10    TI = 1; /* TI: set TI to send first char of UART */
11    while (1)
12    {
13        printf ("Hello World ! \n ");
14    }
15 }
16
```
- Command Window:** Shows the output of the program, which is "Hello World !" repeated multiple times.